

Question number	Answer	Additional guidance	Mark
8(a)(i)	A description that includes the following points: • (viral) RNA interacts with (host cell) ribosomes (1) • credit detail of role of (host) tRNA (1) • peptide bonds form between the (adjacent host) amino acids (1)	ACCEPT two tRNAs bind to RNA (at any one time) each tRNA carries a specific amino acid tRNA anticodons bind to RNA codons tRNA binds to RNA with hydrogen bonds DO NOT ACCEPT polypeptide bond / dipeptide bond	(3)

Question number	Answer	Additional guidance	Mark
8(a)(ii)	An answer that includes three of the following points: - inhibitor binds to protease so that it cannot bind to {the (poly)protein / its substrate} (1) - therefore the (polyprotein peptide) bonds will not be broken (1) - therefore {no / fewer} {individual / smaller / component / named component} proteins will be made (1) - therefore the {components / named components} cannot be assembled (into individual viruses) (1)	ACCEPT inhibitor binds to (poly)protein / inhibitor stops protease binding to (poly)protein ACCEPT (individual) proteins will not be separated ACCEPT virus cannot be put together from (poly)protein	(3)

Question number	Answer	Additional guidance	Mark
8(b)	An explanation that includes four of the following points: • if number of platelets were higher blood would clot faster (1) • and this would help to reduce the bleeding (1) • more {white blood cells / appropriately named white blood cell} {to destroy the virus / increase immune response} (1) • credit a role of a named type of white blood cell in the immune response (1) • credit reason why {phagocytes / named phagocyte} can attach (more) viruses (1)	ACCEPT higher rate e.g. macrophages for antigen presentation to T helper cells T helper cells in activating {T killer cells / B cells} T killer cells to destroy host cells infected with virus B cells to differentiate into plasma cells plasma cells to produce antibody to the virus DO NOT ACCEPT B cells produce antibody antibodies destroy virus e.g. opsonisation agglutination antibodies bind macrophage to the virus viruses released from host cells so accessible to phagocytes	(4)

Question number	Answer	Additional guidance	Mark
8(c)	A description that includes two of the following points: • credit advantage of this method being effective (1) • credit advantage of bacteria being passed to future generations (1) • the mosquitoes are not removed from the food chain (1) • more ethical as mosquitoes not harmed (1)	e.g. decreases chance of epidemic fewer people will die less need for {drugs / vaccines} no side effects of drugs e.g. the prevention will spread as the number of mosquitoes infected with the bacteria increase not every {mosquito / mosquito egg} has to be treated saving {time / money} treatment has to be done only once ACCEPT description of a consequence e.g. food webs not disrupted	(2)

